

Chapter 5: The Network Layer–Control Plane

Exercise 1

Consider the network with the following link weights,

	t	u	v	w	x	y	z
t	0	2	4	-	-	7	-
u	2	0	3	3	-	-	-
v	4	3	0	4	3	8	-
w	-	3	4	0	6	-	-
x	-	-	3	6	0	6	8
y	7	-	8	-	6	0	12
z	-	-	-	-	8	12	0

Using Dijkstra's algorithm, and showing your work using a table similar to Table 5.1, do the following:

- a) Compute the shortest path from t to all network nodes.

visited node	^{Source} t	u	v	w	x	y	z
t	0	∞	∞	∞	∞	∞	∞
u		2	4			7	
v			4	5			
w				5	7	7	
x					7		
y						7	15
z							15

Therefore, The shortest path computed as following:

$$t-u: 2$$

$$t-v: 4$$

$$t-w: 5$$

$$t-x: 7$$

$$t-y: 7$$

$$t-z: 15$$

b) Compute the shortest path from u to all network nodes.

visited node	^{source} u	t	v	w	x	y	z
u	0	∞	∞	∞	∞	∞	∞
t		2	3	3			
v			3			9	
w				3	6	9	
x					6		
y						9	14
z							14

Thus, the computed shortest path from u to all nodes:

$$\begin{aligned}
 u-t &: 2 & u-v-x &: 6 \\
 u-v &: 3 & u-t-y &: 9 \\
 u-w &: 3 & u-v-x-z &: 14
 \end{aligned}$$

c) Compute the shortest path from v to all network nodes.

visited node	^{source} v	t	u	w	x	y	z
v	0	∞	∞	∞	∞	∞	∞
u		4	3	4	3	8	
t		4		4			
y						8	
x					3		20
w				4			11
z							11

Therefore, the computed path from v to all network nodes as follows:

$$\begin{aligned}
 v-u &: 3 & v-y &: 8 & v-w &: 4 \\
 v-t &: 4 & v-x &: 3 & v-x-z &: 11
 \end{aligned}$$

d) Compute the shortest path from w to all network nodes.

visited node	^{source} w	v	t	u	x	y	z
w	0	∞	∞	∞	∞	∞	∞
u		4		3	6		
v		4	5				
t			5		6	12	
y						12	
x					6		24
z							14

Therefore, the computed path from w to all network nodes are:

$$\begin{aligned}
 w-u &: 3 & w-x-y &: 12 \\
 w-v &: 4 & w-x &: 6 \\
 w-ut &: 5 & w-x-z &: 14
 \end{aligned}$$

e) Compute the shortest path from y to all network nodes.

visited node	Source node	y	w	v	t	u	x	z
y	0	∞	∞	∞	∞	∞	∞	∞
x				8	7		6	12
v			12	8				12
t			12		7	11		
u						9		
w			12					
z								12

Thus, the computed shortest path from y to all network nodes are:

$$\begin{aligned}
 y-x &: 6 & y-t &: 7 & y-t-u-w &: 12 \\
 y-v &: 8 & y-t-u &: 9 & y-z &: 12
 \end{aligned}$$

f) Compute the shortest path from z to all network nodes.

visited node	Source node	y	w	v	t	u	x
z	0	∞	∞	∞	∞	∞	8
x		12					
v		12	14				
y		12	14		15	14	
t					15		
u						14	
w			14				

Therefore, the computed path from z to all network nodes:

$$\begin{aligned}
 z-x &: 8 & z-y &: 12 & z-x-w &: 14 \\
 z-x-v &: 11 & z-x-v-t &: 15 & z-x-v-u &: 14
 \end{aligned}$$

Exercise 2

Consider the count-to-infinity problem in the distance vector routing. Will the count-to-infinity problem occur if we decrease the cost of a link? Why? How about if we connect two nodes which do not have a link?

NO, the count-to-infinity problem does not occur when a link cost decreases, because when a link cost decreases, routers immediately prefer lower-cost path, also distance vector routing updates propagated and quickly, so there's no issue of old information persisting. Since routes always choose the shortest path, they immediately switch to the lower-cost route without infinite loop.

NO, the problem does not occur when connecting two nodes, because when a new link is added, the routing table gets updated normally with a new, lower-cost route.